

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai |

Accredited by NAAC with 'A' Grade

Accredited by NBA (B.Tech – IT, B.E – CSE, Civil, MECH, EEE, ECE) (Tier-1)

NAGAPATTINAM – 611 002



B.TECH - INFORMATION TECHNOLOGY

R-2023

Total Credits: 165 (Lateral Entry:125)

FIRST YEAR

SEMESTER I							
COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	Contact Hours
Theory Courses							
2301IP101	Induction Programme	-	-	-	-	-	-
2301MA104	Calculus and Linear Algebra	BSC	3	1	0	4	4
2301GEX01	Foundation of Electrical and Electronics Engineering	ESC	3	0	0	3	3
2301GEX02	Engineering Graphics	ESC	2	1	0	3	3
2301GEX05	Applied Digital Logic and Design	ESC	3	0	0	3	3
	Language Elective	EEC	2	0	0	2	2
2301TA101	Heritage of Tamils / தமிழர் மரபு	HSMC	1	0	0	1	1
Laboratory Courses							
2301GEX51	Computer Practices Laboratory	ESC	0	0	2	1	3
2301GEX53	Foundation of Electrical and Electronics Engineering Laboratory	ESC	0	0	2	1	3
2301GEX54	Applied Digital Logic and Design Laboratory	ESC	0	0	2	1	3
Mandatory Course							
2301LS101	Life Skills - I	-	-	-	-	-	2
TOTAL			14	2	6	19	26

SEMESTER II							
COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	Contact Hours
Theory Courses							
2301MA204	Probability and Statistics	BSC	3	1	0	4	4
2301PH203	Physics for Information Science	BSC	3	0	0	3	3
2301CH201	Applied Chemistry in Informatics	BSC	3	0	0	3	3
2301GEX03	Problem Solving using C	ESC	2	0	4	4	6
2301ENX01	Professional English	HSMC	2	0	0	2	2
2301TA201	Tamils and Technology / தமிழரும் தொழில்நுட்பமும்	HSMC	1	0	0	1	1
Laboratory Courses							
2301PHX51	Engineering Physics Laboratory	BSC	0	0	2	1	3

2301CHX51	Engineering Chemistry Laboratory	BSC	0	0	2	1	3
2301ENX51	Communication Skills Laboratory	HSMC	0	0	2	1	3
2301GEX52	Engineering Practices Laboratory	ESC	0	0	4	2	4
Non Credit / Mandatory Course							
2301LS201	Life Skills - II	-	-	-	-	-	2
TOTAL			14	1	14	22	32

SECOND YEAR

SEMESTER III							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
Theory Courses							
2301MA306	Queuing and Network Model	BSC	3	1	0	4	4
2302IT301	Data Structures and Algorithms	PCC	3	0	0	3	3
2302IT302	Computer Organization and Architecture	PCC	3	0	0	3	3
2302IT303	Problem solving using Python	PCC	2	0	4	4	6
2302IT304	Database Management Systems	PCC	3	0	0	3	3
2301HSX01	Universal Human Values and Ethics	HSMC	1	0	2	2	3
Laboratory Courses							
2302IT351	Data Structures and Algorithms Laboratory	PCC	0	0	2	1	3
2302IT352	Database Management Systems Laboratory	PCC	0	0	2	1	3
2304GE301	Professional Development Course I	EEC	0	0	2	1	2
Non Credit / Mandatory Course							
2301LS301	Life skill III	MC	0	0	0	0	0
Total			14	1	12	22	30

SEMESTER IV							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
Theory Courses							
2302IT401	Java Programming	PCC	2	0	4	4	6
2302IT402	Operating Systems	PCC	3	0	0	3	3
2302IT403	Software Engineering and Project Management	PCC	3	0	0	3	3
2302IT404	Datawarehousing and Data Mining	PCC	3	0	2	4	5
2302IT405	Computer Networks	PCC	3	0	0	3	3
Laboratory Courses							
2302IT451	Operating Systems Laboratory	PCC	0	0	2	1	2
2302IT452	Computer Networks Laboratory	PCC	0	0	2	1	2
2301GEX07	Environmental Sciences and Sustainability	BSC	1	0	2	2	3
2304GE401	Professional Development Course II	EEC	0	0	2	1	2
Non Credit / Mandatory Course							
2304MC451	Mandatory Course I	MC	3	0	0	0	3
2301LS401	Life skill IV	MC	0	0	0	0	0
Total			18	0	12	22	32

THIRD YEAR

SEMESTER V							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
Theory Courses							
2302IT501	Object Oriented Analysis and Design	PCC	3	0	0	3	3
2302IT502	Web Programming	PCC	3	0	0	3	3

2302IT503	Mobile Computing	PCC	3	0	0	3	3
2302IT504	Artificial Intelligence and Machine Learning	PCC	3	0	2	4	5
	Professional Elective I	PEC	3	0	0	3	3
	Open Elective I	OEC	3	0	0	3	3
Laboratory Courses							
2302IT551	Web Programing Lab	PCC	0	0	2	1	3
2304IT551	CASE Tools Lab(Mini Project I)	EEC	0	0	2	1	3
2301GE501	Professional Development Course III	EEC	0	0	2	1	2
2304IT552	Startup Opportunities for IT Engineers	EEC	2	0	0	2	2
Non Credit / Mandatory Course							
2301LS501	Life skill V	MC	0	0	0	0	0
Total			20	0	8	24	28

SEMESTER VI							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
Theory Courses							
2302IT601	Full Stack Development	PCC	2	0	4	4	6
2302IT602	Foundations of Cryptography	PCC	3	0	2	4	6
2302IT603	Embedded systems and IoT	PCC	3	0	0	3	3
	Professional Elective II	PEC	3	0	0	3	3
	Professional Elective III	PEC	3	0	0	3	3
	Open Elective II	OEC	3	0	0	3	3
Laboratory Courses							
2302IT651	Embedded systems and IoT Lab	PCC	0	0	2	1	3
2304IT651	Software Development (Mini Project II)	EEC	0	0	4	2	4
2301GE601	Professional Development Course IV	EEC	0	0	2	1	2
Non Credit / Mandatory Course							
2304MC651	Mandatory Course II	MC	3	0	0	0	3
2301LS601	Life skill VI	MC	0	0	0	0	0
Total			21	0	12	24	36

FINAL YEAR

SEMESTER VII							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
Theory Courses							
2302IT701	Foundation Models	PCC	3	0	2	4	5
2302IT702	Cloud Computing	PCC	3	0	2	4	5
	Professional Elective IV	PEC	3	0	0	3	3
	Professional Elective V	PEC	3	0	0	3	3
	Professional Elective VI	PEC	3	0	0	3	3
	Open Elective III	OEC	3	0	0	3	3
Laboratory Courses							
2304IT751	Project Work Phase I	EEC	0	0	4	2	4
	Industry Certification	EEC	0	0	0	1	0
Total			17	0	10	23	26

SEMESTER VIII							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
Laboratory Courses							
	Internship Presentation	EEC	0	0	0	1	0
2304IT851	Project Work (Phase II)	EEC	0	0	20	8	20
Total			0	0	0	9	20

PROFESSIONAL ELECTIVE COURSES: VERTICALS

VERTICAL I : DATA SCIENCE						
Course Code	Course Name	L	T	P	C	Contact Hours
2303IT001	Exploratory Data Analysis	3	0	0	3	3
2302IT702	Computer Vision	3	0	0	3	3
2303IT003	Big Data	3	0	0	3	3
2303IT004	Text and Speech Analysis	3	0	0	3	3
2303IT005	Foundations of Data Science	3	0	0	3	3
2303IT006	Business Analytics	3	0	0	3	3

VERTICAL 2: INTELLIGENT COMPUTING						
Course Code	Course Name	L	T	P	C	Contact Hours
2303IT007	Knowledge Engineering	3	0	0	3	3
2302IT708	Neural Networks And Deep Learning	3	0	0	3	3
2303IT009	Soft Computing	3	0	0	3	3
2303IT010	Social Network Analysis	3	0	0	3	3
2303IT011	Cognitive Science	3	0	0	3	3
2303IT012	Generative AI	3	0	0	3	3

VERTICAL 3: WEB DESIGN AND DEVELOPMENT						
Course Code	Course Name	L	T	P	C	Contact Hours
2303IT013	C# and .NET	3	0	0	3	3
2303IT014	Devops	3	0	0	3	3
2303IT015	UI and UX Design	3	0	0	3	3
2303IT016	MERN Stack Development	3	0	0	3	3
2303IT017	Web Application Security	3	0	0	3	3
2303IT018	Digital Marketing	3	0	0	3	3

VERTICAL 4: CYBER SECURITY						
Course Code	Course Name Category	L	T	P	C	Contact Hours
2303IT019	Modern Cryptography	3	0	0	3	3
2303IT020	Digital and Mobile Forensics	3	0	0	3	3
2303IT021	Social Network Security	3	0	0	3	3
2303IT022	Engineering Secure Software Systems	3	0	0	3	3
2303IT023	Ethical Hacking	3	0	0	3	3
2303IT024	Cyber Security	3	0	0	3	3

VERTICAL 5: COMPUTATIONAL SYSTEMS						
Course Code	Course Name	L	T	P	C	Contact Hours
2303IT025	Data Centre and Virtualization	3	0	0	3	3
2303IT026	Green Computing	3	0	0	3	3
2303IT027	Quantum Computing	3	0	0	3	3
2303IT028	High Performance Computing	3	0	0	3	3
2303IT029	Natural Language Processing	3	0	0	3	3
2303IT030	Serverless Computing	3	0	0	3	3

VERTICAL 6: EMERGING TECHNOLOGIES						
Course Code	Course Name	L	T	P	C	Contact Hours
2303IT031	Building Smart Cities	3	0	0	3	3
2303IT032	Game Programming	3	0	0	3	3
2303IT033	Block Chain Technologies	3	0	0	3	3
2303IT034	Salesforce Business Application	3	0	0	3	3
2303IT035	Augmented Reality/Virtual Reality	3	0	0	3	3
2303IT036	3D Design and Printing	3	0	0	3	3

OPEN ELECTIVE COURSES OFFERED TO OTHER DEPARTMENTS (OE)

OPEN ELECTIVE I							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
2303IT037	Datawarehousing and Data mining	OEC	3	0	0	3	3
2303IT038	Multimedia systems and Its Applications	OEC	3	0	0	3	3
2303IT039	Object Oriented Programming using C++	OEC	3	0	2	4	5
2303IT040	Management Information System	OEC	3	0	0	3	3

OPEN ELECTIVE II							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
2303IT041	AI for Everyone	OEC	3	0	0	3	3
2303IT042	Information Extraction and Text Mining	OEC	3	0	0	3	3
2303IT043	Green Computing	OEC	3	0	0	3	3
2303IT044	Speech Processing	OEC	3	0	0	3	3

OPEN ELECTIVE II							
Course Code	Course Name	Category	L	T	P	C	Contact Hours
2303IT045	Fundamentals of Cloud Computing	OEC	3	0	0	3	3
2303IT046	Introduction to Ethical Hacking	OEC	3	0	0	3	3
2303IT047	Internet of Things	OEC	3	0	0	3	3
2303IT048	Mobile Application Development	OEC	3	0	0	3	3

Mandatory Course-I (R-2023)

Course Code	Course Name
2301MC401	Design Thinking for Innovation
2301MC402	Circular Economy
2301MC403	Innovation & Entrepreneurship Fundamentals
2301MC404	Startup Essentials
2301MC405	Lean Start-Up Methodology
2301MC406	Social Entrepreneurship
2301MC407	Business Model Innovation
2301MC408	Intellectual Property Rights for Engineers
2301MC409	Sustainable Development Goals (SDGs) for Engineers

Mandatory Course-II

Course Code	Course Name
2301MC601	Introduction to Women and Gender Studies
2301MC602	Elements of Literature
2301MC603	Film Appreciation
2301MC604	Disaster Risk Reduction and Management
2301MC605	Well-Being with Traditional Practices-Yoga, Ayurveda and Siddha
2301MC606	History of Science and Technology in India
2301MC607	Political and Economic Thought for a Humane Society
2301MC608	State, Nation Building and Politics in India
2301MC609	Industrial Safety
2301MC610	Stress Management by Yoga
2301MC611	Constitution of India
2301MC612	Essence of Indian Traditional Knowledge
2301MC613	Mental Health & Wellbeing

*Mandatory courses are to be offered as Non-Credit courses.

SUMMARY

Name of the Programme : B.Tech - Information Technology											
S.No	Category	Semester									%
		I	II	III	IV	V	VI	VII	VIII	Total	
1	HSMC	1	4	2	-	-	-	-	-	7	4.2
2	BSC	4	12	4	2	-	-	-	-	22	13.3
3	ESC	12	6	-	-	-	-	-	-	18	10.9
4	PCC	-	-	15	19	14	12	8	-	68	41.3
5	PEC	-	-	-	-	3	6	9	-	18	10.9
6	OEC	-	-	-	-	3	3	3	-	9	5.5
7	EEC	2	-	1	1	4	3	3	9	23	13.9
8	Non-Credit /(Mandatory)	√	√	√	√	√	√	-	-	-	-
Total		19	22	22	22	24	24	23	9	165	100.0

ENROLLMENT FOR B.E. / B. TECH. (HONOURS) / MINOR DEGREE (OPTIONAL)

A student can also optionally register for additional courses (18 credits) and become eligible for the award of B.E. / B. Tech. (Honours) or Minor Degree.

For B.E. / B. Tech. (Honours), a student shall register for the additional courses (18 credits) from semester V onwards. These courses shall be from the same vertical or a combination of different verticals of the same programme of study only.

For minor degree, a student shall register for the additional courses (18 credits) from semester V onwards. All these courses have to be in a particular vertical from any one of the other programmes, Moreover, for minor degree the student can register for courses from any one of the following verticals also.

Complete details are available in clause 4.10 of Regulations 2021 (Anna University).

B.E./B.Tech. Minor with specialisation in another discipline
(In addition to all the verticals of other programmes)

CYBER SECURITY

Minor Degree in CYBER SECURITY										
COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS			
							CA	ES	TOTAL	
Theory Courses										
	Introduction to Cyber Security	PCC	3	0	0	3	40	60	100	
	Network and Web Security	PCC	3	0	0	3	40	60	100	
	Applied Cryptography and Secure Protocols	PCC	3	0	0	3	40	60	100	
	Cloud and IoT Security	PCC	3	0	0	3	40	60	100	
Theory Combined Laboratory Courses										
	Ethical Hacking	PCC	2	0	2	4	50	50	100	
Elective Courses										
	Digital Forensics & Incident Response	PEC	3	0	0	3	40	60	100	
	Cyber Risk & Governance	PEC	3	0	0	3	40	60	100	
	Malware Analysis and Reverse Engineering	PEC	3	0	0	3	40	60	100	
TOTAL			17	0	2	19	250	350	600	

One Credit/Value Added/Diploma Courses:

One / Two Credit Courses:	Value Added / Add-on Courses	Diploma Certification Courses
1. Oracle Java 2. Oracle PL/SQL 3. NodeJS 4. ReactJS 5. AWS 6. Salesforce Trail Head 7. Dell EMC Data Scientist 8. Palo Alto Cyber Security Administrator 9. UiPath Automation 10. Ethical Hacking	1. Infosys Springboard 2. Skillrack and EBox 3. Top Freshers 4. TCS iON 5. Edu Skills 6. Mooc Courses 7. Other Industry Supported/ Need based Courses	1. Programming and Problem Solving 2. Hardware and Trouble Shooting 3. Mobile App Development 4. Web Designing 5. Animation and Photoshop